

Medical Clinic Management System

Database Project — Phase 1

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1. System Objectives

The Medical Clinic Management System aims to provide a structured and efficient database solution for managing the core operations of a medical clinic. The system is designed to achieve the following objectives:

- Store and manage complete doctor profiles including their supervisory hierarchy.
- Maintain patient records with personal details and contact information.
- Track appointments scheduled between patients and doctors.
- Record prescriptions issued by doctors to patients.
- Manage the catalog of medications and their inclusion in prescriptions.

2. System Requirements

The following requirements describe the data that the system must capture and the rules that govern the relationships between entities:

Doctors

The clinic employs a number of doctors. Each doctor is identified by a unique Doctor ID and has a composite name (first, middle, last), a specialization, and a phone number. Within the clinic, a senior doctor may supervise one or more other doctors, forming a recursive supervisory relationship among the medical staff.

Patients

The clinic serves patients, each identified by a unique Patient ID. Every patient has a composite name (first, middle, last), a date of birth, and may have multiple home addresses recorded in the system.

Appointments

A patient may book appointments with one or more doctors, and a doctor may have appointments with many patients. Each appointment is characterized by a date and time, which serve as attributes of the appointment relationship between a patient and a doctor.

Prescriptions

During a consultation, a doctor may issue a prescription for a patient. Each prescription has a unique prescription number and an issue date. A prescription is associated with exactly one doctor and one patient.

Medications

Each prescription may contain one or more medications. A medication has a unique medication name, a type, and a dosage. The same medication can appear in many different prescriptions, forming a many-to-many relationship between prescriptions and medications.

3. Raw Scenario

Scenario: Medical Clinic

The clinic employs a number of Doctors. Each doctor has a unique doctor ID, a composite name (first, middle, last), a specialization, and a phone number. Doctors can supervise other doctors within the clinic (a recursive relationship — e.g., a senior doctor supervises residents).

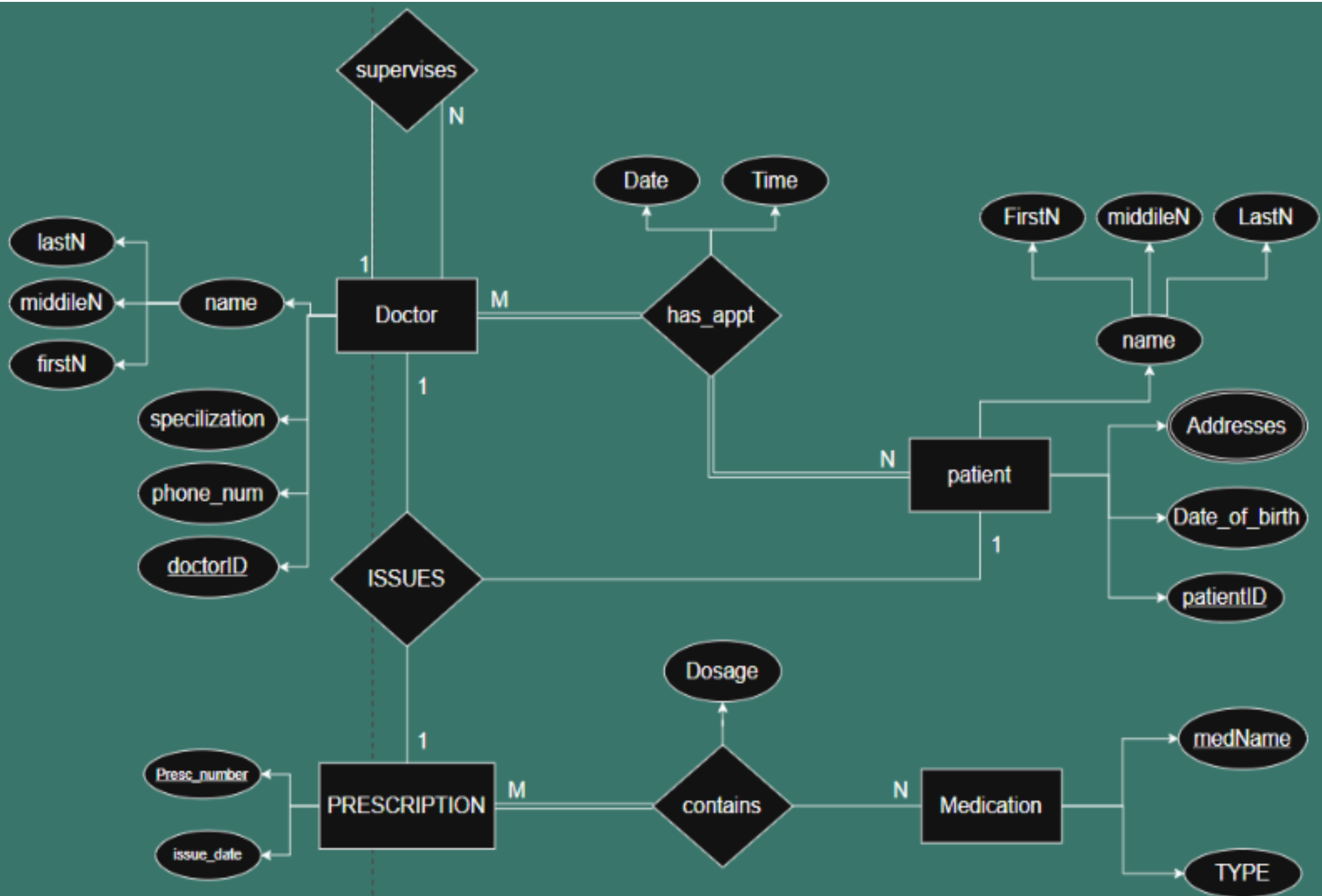
The clinic serves Patients. Each patient has a unique patient ID, a composite name (first, middle, last), a date of birth, and multiple addresses (multivalued attribute).

Each patient books Appointments with a doctor. An appointment has a date and time. A patient must have at least one appointment to exist in the system — meaning Appointment is a weak entity identified by its date+time combined with the patient it belongs to.

During an appointment, the doctor may issue a Prescription. Each prescription has a unique prescription number, and a date. A prescription contains one or more Medications, and each medication has a name, dosage, and type. The same medication can appear in many prescriptions (M:N).

4. ER Diagram

The ER diagram on the following page represents the conceptual schema of the Medical Clinic Management System using standard ER notation as covered in Chapter 3.



5-MAPPING

